

THE MOD-2 COKERNEL DENSITY OF A HECKE OPERATOR ON A NON-ANALYTIC PRO-2 TOWER: SPLIT FORM, EXPLICIT KERNELS, AND BOUNDS

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ABSTRACT. Let $P = (1 + \varphi\mathcal{O})/(1 + t\mathbb{F}_2[[t]])$ be the pro-2 group attached by R. Chen to the Cartwright–Mantero–Steger–Zappa $\tilde{A}_2$ -lattice of order $q = 2$, where $\mathcal{O} = \mathbb{F}_8[[\varphi; \sigma]]$, $\varphi^3 = t$, and let $F \in \mathbb{F}_2[\Gamma^0]$ be the element whose right-multiplication cokernel computes the $\mathbb{F}_2$ -corank of the Hecke Laplacian along the canonical tower. Chen asked whether the cokernel density c_e vanishes, and whether the restricted enveloping algebra of $\mathrm{gr} P$ is a domain. We answer the second question negatively, exhibiting an explicit square-zero element in degree 4. We show that $\mathrm{gr} P \otimes \mathbb{F}_8$ is the positive part of the affine Lie algebra $A_2^{(1)}$ in its principal grading, so that $\mathrm{gr} P$ is a twisted $\mathbb{F}_2$ -form of it, and that the leading form of F is homogeneous of weight equal to the null root δ . This yields explicit kernel elements for every level (governed by a “binary odometer” along the imaginary root chain $3 \cdot 2^r$), monotonicity statements, and, by exact computation up to $|G_{10}| = 2^{24}$, the bound $c_e \leq 0.030322$ (previously 0.0645). Using evaluation representations of the loop algebra we prove that the leading form of F is a non-zero-divisor in $\mathrm{gr} \mathbb{F}_2[[P]]$, hence that F is a non-zero-divisor in $\mathbb{F}_2[[P]]$. Finally we prove that $c_e = 0$ follows from two explicit hypotheses on the kernel of the leading form (verified for $m \leq 10$ and $m \leq 14$ respectively), we prove that the natural level-by-level route to these hypotheses cannot succeed, and we give computational evidence that the zero divisors of $\mathrm{gr} \mathbb{F}_2[[P]]$ do not lift to $\mathbb{F}_2[[P]]$, consistent with Jaikin-Zapirain’s pro- p Atiyah conjecture. In a last section we treat the analogous graded algebras for all cyclic algebras of degree ℓ over $\mathbb{F}_p((t))$: the non-zero-divisor theorem holds whenever $p \nmid \ell$, the explicit kernels exist only for $(p, \ell) = (2, 3)$ (because of a factor 2 in a characteristic-0 obstruction) and for $\ell = 2$, and for $\ell = 2$ we find a second explicit kernel family, coming from the substitution $t \mapsto t^2$, and evidence that the excess over the maximal-rank baseline vanishes.

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1. INTRODUCTION

1.1. The question. Let $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$ with $\alpha^3 = \alpha + 1$, $\sigma(a) = a^2$, and let D be the cyclic division algebra $(\mathbb{F}_8(t)/\mathbb{F}_2(t), \sigma, t)$, i.e. $D = \bigoplus_{j < 3} \mathbb{F}_8(t)\varphi^j$ with $\varphi a = \sigma(a)\varphi$ and $\varphi^3 = t$. Chen [1, Thm. 6.4.1] shows that $a_x \mapsto 1 + \alpha^x \varphi$ ($x \in \mathbb{Z}/7$) embeds the Cartwright–Mantero–Steger–Zappa triangle group Γ of order $q = 2$ [2] into $D^\times/\mathbb{F}_2(t)^\times$, so that Γ is a lattice in $PGL_3(\mathbb{F}_2((s)))$, $s = 1 + t$. Completing at $t = 0$, with maximal order $\mathcal{O} = \mathbb{F}_8[[\varphi; \sigma]]$, one obtains the pro-2 group

$$P = (1 + \varphi\mathcal{O})/(1 + t\mathbb{F}_2[[t]]), \quad P_m = \text{im}(1 + \varphi^m\mathcal{O}), \quad G_m = P/P_m,$$

with $|G_m| = 2^{N_m}$, $N_m = 3(m-1) - \lfloor (m-1)/3 \rfloor$. The type-preserving subgroup Γ^0 is dense in P ; P is torsion-free, $d(P) = 3$, of infinite rank and not p -adic analytic [1, Thm. 6.4.2].

With Schreier generators $\theta_{s,x}$ and $\bar{f}_s = \sum_{x \in \mathbb{Z}/7} \theta_{s,x}^{-1} \in \mathbb{F}_2[G_m]$ put

$$F = 1 + \bar{f}_1 \bar{f}_0 \bar{f}_2, \quad C_m = \mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F, \quad n_m = 3|G_m|.$$

Then $\dim C_m - 1$ is the number of even invariant factors of the Hecke Laplacian of the level- m quotient complex [1, Prop. 6.8.3], $\dim C_m/n_m$ is non-increasing, and

$$c_e := \lim_{m \rightarrow \infty} \dim C_m/n_m$$

controls the linear growth of the 2-part of the Hecke Jacobian: $\liminf v_2 |\text{Jac}^H(Y_m)|/n_m \geq c_e$ [1, Thm. 6.8.8]. Chen asks whether $c_e = 0$, and (Remark 6.9.2 of [1]) whether the restricted enveloping algebra of $\text{gr } P$ is a domain, which would make $\mathbb{F}_2[[P]]$ a domain.

1.2. Relation to the pro- p Atiyah conjecture. By Jaikin-Zapirain [3, Prop. 11.2], for any pro- p group G and any chain of open normal subgroups with trivial intersection, $\text{rk}_G(x) = \lim \text{rk}_{\mathbb{F}_p}(x \text{ on } \mathbb{F}_p[G/U_i])/|G : U_i|$ exists and does not depend on the chain. Hence

$$c_e = \frac{1}{3}(1 - \text{rk}_P(F))$$

is intrinsic to (P, F) . The pro- p Atiyah conjecture [3, Conj. 11.1] for the torsion-free group P would give $\text{rk}_P(F) = 1$, i.e. $c_e = 0$. It is known for residually torsion-free p -adic analytic groups, free pro- p groups and virtually free-by- $\mathbb{Z}_p$ groups [4], none of which covers P . Note that the reduced norm maps the central subgroup $1 + t\mathbb{F}_2[[t]]$ onto itself bijectively (it is cubing on a pro-2 group), so that $P \cong SL_1^1(\hat{D})$, the first principal congruence subgroup of the norm-one group of $\hat{D} = D \otimes_{\mathbb{F}_2(t)} \mathbb{F}_2((t))$; P is thus an anisotropic analytic group over the local field $\mathbb{F}_2((t))$, a class for which the pro- p Atiyah conjecture is open. For the analogous questions on the growth of mod- p homology along towers see [11, 12].

1.3. Main results. Write $L = \text{gr } P = \bigoplus_{k \geq 1} L_k$ for the graded restricted Lie algebra of the filtration (P_k) , $u(L)$ for its restricted enveloping algebra and $u_m = u(L/L_{\geq m}) \cong \text{gr } \mathbb{F}_2[G_m]$ (Jennings–Quillen). Let $\sigma F \in u(L)_3$ be the leading form of F .

Theorem A (Section 3). *$u(L)$ is not a domain. There is a 3-dimensional subspace $V \subset u(L)_4$ with $V \cdot V = 0$; in particular every $0 \neq v \in V$ satisfies $v^2 = 0$. There are no homogeneous zero-divisor pairs of total degree ≤ 7 . Hence the answer to Remark 6.9.2 of [1] is negative.*

Theorem B (Section 4). *$L \otimes_{\mathbb{F}_2} \mathbb{F}_8$ is isomorphic, as a graded restricted Lie algebra, to the positive part $\widehat{\mathfrak{n}}$ of the affine Lie algebra of type $A_2^{(1)}$ in its principal grading, with 2-map given by matrix squaring. L is the $\mathbb{F}_2$ -form of $\widehat{\mathfrak{n}}$ twisted by Frobenius composed with the rotation of the affine Dynkin diagram. Under this isomorphism the leading form σF becomes*

$$\sigma' = e_0 e_1 e_2 + e_2[e_0, e_1] + e_0[e_1, e_2] + h, \quad h \in \widehat{\mathfrak{n}}_\delta,$$

which is homogeneous of weight $\delta = \alpha_0 + \alpha_1 + \alpha_2$ for the affine root grading.

Theorem C (Sections 5–6). *For every $m \geq 2$ there are three explicit monomials $\omega_0, \omega_1, \omega_2 \in u_m \otimes \mathbb{F}_8 \cong u_m^s \otimes \mathbb{F}_8$ of degree*

$$d(m) = \sum_{k \in \mathcal{H}_m} k, \quad \mathcal{H}_m = \{1 \leq k < m : 3 \nmid k\} \cup \{3 \cdot 2^r < m\},$$

such that $u_m^s \omega_0 + u_m^s \omega_1 + u_m^s \omega_2$ lies in the kernel of right multiplication by σF . Moreover the first non-injective degree k_m of $R_{\sigma F}$ is non-decreasing in m , the kernel density $\dim \ker R_{\sigma F} / |G_m|$ is non-increasing in m , and $k_m \leq d(m) \sim m^2/3$.

Theorem D (Section 8; exact computation). *$\dim C_7 = 10226$, and the graded cokernel of $R_{\sigma F}$ on u_m has dimension 64216, 423412, 1526176 for $m = 8, 9, 10$. Consequently*

$$c_e \leq \frac{1526176}{3 \cdot 2^{24}} = 0.030322 \dots$$

In Section 7 we prove that right and left multiplication by the leading form are injective on the full algebra $u(\widehat{\mathfrak{n}})$ (Theorem 7.1); thus F is a non-zero-divisor in $\mathbb{F}_2[[P]]$ and $k_m \geq m - 3$. In Section 9 we split the kernel density as a maximal-rank baseline, which we show is $O(m^{-1/2})$ (numerically $\approx 2.55 m^{-3/2}$), plus an excess e_m of order 10^{-3} , and formulate $c_e = 0$ as a Lefschetz-type statement (Conjecture 9.2). We prove that $c_e = 0$ follows from a growth bound on the kernel below the middle degree together with $k_m \geq d(m) - 1$ (Theorem 9.4); both hypotheses hold for $m \leq 10$, and we show that the excess of σ' is a special-position effect: a generic element of the same weight has maximal rank on every block for $m \leq 8$. In Section 10 we show by exhaustive linear algebra that every zero-divisor pair of Theorem A is obstructed at the first or second lifting step in $\mathbb{F}_2[[P]]$; thus they are artefacts of the filtration and give no counterexample to the pro-2 Kaplansky conjecture. In Section 11 we compare with the other cyclic algebras: Theorem 11.1 extends the non-zero-divisor theorem to all $p \nmid \ell$, Proposition 11.3 shows that the column mechanism behind Theorem C is special to $(2, 3)$ and $\ell = 2$, and Theorem 11.4 gives two explicit kernel families for $\ell = 2$, which together with occasional cancellation-type defects account for all computed thresholds there.

Status of the statements. Results labelled *Theorem*, *Proposition* or *Lemma* have complete proofs here, possibly relying on a finite exact computation which is then named explicitly (“proof by computation”). All computations are exact linear algebra over $\mathbb{F}_2$ (over $\mathbb{F}_p$ in Section 11) and were cross-checked by independent implementations; see Appendix A.

2. THE GRADED ALGEBRA

2.1. The Lie algebra L . Let $A = \mathbb{F}_8[\varphi; \sigma]$ (graded by $\deg \varphi = 1$); its centre is $\mathbb{F}_2[t]$, $t = \varphi^3$.

Lemma 2.1. *The filtration (P_k) is a restricted N -series and*

$$L \cong A_{>0}/T, \quad T = t\mathbb{F}_2[t],$$

as graded restricted Lie algebras, where $A_{>0}$ carries the commutator bracket and the squaring map. Explicitly, $L_k = \mathbb{F}_8\varphi^k$ for $3 \nmid k$ and $L_k = \mathbb{F}_8\varphi^k/\mathbb{F}_2t^{k/3}$ for $3 \mid k$, and

$$[a\varphi^i, b\varphi^j] = (a\sigma^i(b) + b\sigma^j(a))\varphi^{i+j}, \quad (a\varphi^k)^{[2]} = a\sigma^k(a)\varphi^{2k}.$$

Proof. For $x = 1 + a\varphi^i + \dots$, $y = 1 + b\varphi^j + \dots$ one has $[x, y] \equiv 1 + (a\sigma^i(b) + b\sigma^j(a))\varphi^{i+j}$ and $x^2 \equiv 1 + a\sigma^i(a)\varphi^{2i}$ modulo higher order; these are the commutator and square in A . The centre T is closed under squaring, hence a restricted ideal, and quotienting by $1 + t\mathbb{F}_2[[t]]$ removes exactly it. $\square$

Write $d_k = \dim_{\mathbb{F}_2} L_k$, so $d_k = 3$ for $3 \nmid k$ and $d_k = 2$ for $3 \mid k$. In particular $d_{2k} = d_k$.

Proposition 2.2. *The Hilbert series of $u(L)$ is*

$$\sum_j \dim u(L)_j z^j = \prod_{k \geq 1} (1 + z^k)^{d_k} = \prod_{k \text{ odd}} (1 - z^k)^{-d_k}.$$

In particular $u(L)$ has the Hilbert series of a commutative polynomial ring with d_k generators in each odd degree k , and has subexponential growth.

Proof. The first equality is PBW for restricted enveloping algebras. For the second use $d_{2^j k} = d_k$ and $\prod_{j \geq 0} (1 + x^{2^j}) = (1 - x)^{-1}$. $\square$

2.2. The leading form of F . By Jennings–Quillen [5, 6], $\text{gr}_I \mathbb{F}_2[G_m] \cong u(L/L_{\geq m}) =: u_m$, where I is the augmentation ideal; in degrees $< m$, u_m agrees with $u(L)$.

Proposition 2.3 (proof by computation). *$F \in I^3 \setminus I^4$, and its class $\sigma F \in u(L)_3$ is, independently of $m \geq 4$,*

$$\sigma F = x_{1,1}x_{1,\alpha}x_{1,\alpha^2} + x_{1,1}x_{2,\alpha} + x_{1,\alpha^2}x_{2,\alpha} + x_{1,1}x_{2,\alpha^2} + x_{1,\alpha}x_{2,\alpha^2},$$

where $x_{k,b}$ is the class of $1 + b\varphi^k$ and PBW monomials are ordered by (k, b) .

Lemma 2.4 (semicontinuity). *For every m , $\dim C_m \leq \dim \text{coker}(R_{\sigma F} : u_m \rightarrow u_m)$, where $R_{\sigma F}(y) = y\sigma F$. Consequently $c_e \leq \dim \text{coker}(R_{\sigma F}|_{u_m})/n_m$ for every m .*

Proof. Right multiplication by F maps I^j into I^{j+3} and induces $R_{\sigma F}$ on the associated graded. For a filtered map f of finite-dimensional filtered spaces, $\text{rk } f \geq \text{rk gr } f$. The second claim follows from the monotonicity of $\dim C_m/n_m$ [1, Prop. 6.8.3]. $\square$

3. $u(L)$ IS NOT A DOMAIN

Theorem 3.1 (proof by computation). *Let*

$$\begin{aligned} v = & x_{1,1}x_{1,\alpha}x_{2,1} + x_{1,\alpha}x_{1,\alpha^2}x_{2,1} + x_{1,\alpha}x_{1,\alpha^2}x_{2,\alpha} + x_{1,1}x_{1,\alpha}x_{2,\alpha^2} \\ & + x_{1,1}x_{1,\alpha^2}x_{2,\alpha^2} + x_{1,\alpha}x_{1,\alpha^2}x_{2,\alpha^2} + x_{2,1}x_{2,\alpha^2} + x_{2,\alpha}x_{2,\alpha^2} \in u(L)_4. \end{aligned}$$

Then $v \neq 0$ and $v^2 = 0$ in $u(L)_8$. More precisely, there is a 3-dimensional subspace $V \ni v$ of $u(L)_4$ with $V \cdot V = 0$, and the left and right kernels of multiplication by v on $u(L)_4$ are both equal to V . Every homogeneous pair $x, y \neq 0$ with $xy = 0$ has $\deg x + \deg y \geq 8$.

Verification. (i) In a PBW implementation of $u(L)$ (Lemma 2.1), checked against the Hilbert series, associativity on random triples, and the group algebra (Jennings products in $\mathbb{F}_2[G_7]$ and $\mathbb{F}_2[G_{12}]$), $v^2 = 0$ and $V \cdot V = 0$. (ii) Independently of that code: lifting v to an element $\tilde{v} \in \mathbb{F}_2[G_9]$ supported on 18 group elements, the product $\tilde{v} \tilde{w}$ ($w \in V$) has no Jennings coordinates of weight ≤ 8 , so $\tilde{v} \tilde{w} \in I^9$, which by Jennings' basis theorem is the statement $vw = 0$ in $\text{gr}_8 \mathbb{F}_2[G_9] = u(L)_8$. (iii) The last assertion is an exhaustive rank computation: for all nonzero homogeneous x of degree $a \leq 3$ and the stated ranges of b (degree 1: $b \leq 23$; degree 2: $b \leq 20$; degree 3: $b \leq 13$; degree 4: $b \leq 3$) left and right multiplication $u_b \rightarrow u_{a+b}$ is injective. $\square$

Remark 3.2. A second family exists in degrees $(3, 14)$: $x = x_{1,1}x_{2,1} + x_{1,\alpha}x_{2,\alpha} + x_{1,\alpha}x_{2,\alpha^2} + x_{1,\alpha^2}x_{2,\alpha^2}$ annihilates a 3-dimensional subspace of $u(L)_{14}$. By contrast, σF is not a zero divisor in low degrees: left and right multiplication $u_b \rightarrow u_{b+3}$ by σF is injective for all $b \leq 28$. The conceptual source of these zero divisors is explained by Theorem B: after $\otimes \mathbb{F}_8$ the real root vectors square to zero, and L only hides this through its twisted $\mathbb{F}_2$ -structure.

4. THE SPLIT FORM

Let $E = \mathbb{F}_2^3$ with idempotents e_0, e_1, e_2 , $s(e_p) = e_{p+1}$ (indices mod 3), $A^s = E[\varphi; s]$, $T^s = t\mathbb{F}_2[t]$ with $t = (1, 1, 1)\varphi^3$, and $L^s = A^s_{>0}/T^s$.

Proposition 4.1. *The map $\psi : A \otimes_{\mathbb{F}_2} \mathbb{F}_8 \rightarrow A^s \otimes_{\mathbb{F}_2} \mathbb{F}_8$, $a\varphi^k \otimes \lambda \mapsto (\lambda a, \lambda\sigma^{-1}(a), \lambda\sigma^{-2}(a))\varphi^k$, is an isomorphism of graded $\mathbb{F}_8$ -algebras with $\psi(T \otimes \mathbb{F}_8) = T^s \otimes \mathbb{F}_8$. Hence $L \otimes \mathbb{F}_8 \cong L^s \otimes \mathbb{F}_8$, $u(L) \otimes \mathbb{F}_8 \cong u(L^s) \otimes \mathbb{F}_8$ and $u_m \otimes \mathbb{F}_8 \cong u_m^s \otimes \mathbb{F}_8$.*

Proof. In degree 0 this is the classical isomorphism $\mathbb{F}_8 \otimes \mathbb{F}_8 \cong \prod_{\text{Gal}} \mathbb{F}_8$, which intertwines $\sigma \otimes 1$ with the shift s . It therefore extends to the skew polynomial rings with $\varphi \mapsto \varphi$, bijectively in each degree, and $t \mapsto (1, 1, 1)\varphi^3$. Restricted structures are defined by products. $\square$

Proposition 4.2. *Let $\Pi = E_{10} + E_{21} + tE_{02} \in M_3(\mathbb{F}_2[t])$. Then $\sum_k D_k \varphi^k \mapsto \sum_k D_k \Pi^k$ (D_k diagonal) identifies A^s with the Iwahori order $\{X \in M_3(\mathbb{F}_2[t]) : X(0) \text{ lower triangular}\}$, graded by the principal grading $\deg E_{ij}t^n = 3n + (i - j)$, and identifies L^s with the nilradical of the standard Iwahori subalgebra of $\mathfrak{sl}_3 \otimes \mathbb{F}_2[t, t^{-1}]$, i.e. with the positive part $\hat{n}$ of $A_2^{(1)}$ in the principal grading [8, Ch. 7 and 14]. The degree-one elements $e_p\varphi$ are the Chevalley generators; real root vectors square to zero and $(ht^n)^{[2]} = h^2t^{2n}$ for Cartan elements.*

Proof. $\Pi E_{qq} = E_{q+1, q+1}\Pi$ and $\Pi^3 = tI$, so the map is an injective graded algebra homomorphism with the stated image. Since 3 is invertible in $\mathbb{F}_2$, $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus \mathbb{F}_2 I$, and $x \mapsto x + \text{tr}(x)I$ identifies $\mathfrak{gl}_3/\mathbb{F}_2 I$ with $\mathfrak{sl}_3$ as restricted Lie algebras ($\text{tr}(x^2) = \text{tr}(x)^2$ in characteristic 2). The dimension count $d_k = 3$ ($3 \nmid k$: three real roots of principal height k) and $d_k = 2$ ($3 \mid k$: imaginary root of multiplicity rank $\mathfrak{sl}_3$) is consistent. $\square$

4.1. Gradings and symmetries. A^s is the path algebra of the oriented 3-cycle, so L^s , $u(L^s)$ and u_m^s are graded by $Q_+ = \mathbb{N}\beta_0 \oplus \mathbb{N}\beta_1 \oplus \mathbb{N}\beta_2$ (β_p the weight of $e_p\varphi$, the simple roots), refining the principal degree; t has weight $\delta = \beta_0 + \beta_1 + \beta_2$. The rotation $\rho(e_p\varphi^k) = e_{p+1}\varphi^k$ is an automorphism, and $\tau(e_p\varphi^k) = e_{k-p}\varphi^k$ is an anti-automorphism of A^s fixing T^s ; in characteristic 2 it induces a restricted Lie automorphism of L^s and an anti-automorphism Θ' of u_m^s .

Proposition 4.3 (proof by computation for the first identity). *With $y_{k,p} = e_p\varphi^k$,*

$$\sigma' := \psi(\sigma F) = y_{1,0}y_{1,1}y_{1,2} + y_{1,2}y_{2,1} + y_{1,0}y_{2,2} + y_{3,1} = c_1(y_{1,1}y_{1,2} + y_{2,2}) + c_2 y_{1,2} + c_3,$$

where $c_k := e_{k-1}\varphi^k$. In particular σ' has $\mathbb{F}_2$ -coefficients, is Q -homogeneous of weight δ , and $\rho(\sigma') = \Theta'(\sigma') = \sigma'$.

Corollary 4.4. *For all m, n , $\text{rk}(R_{\sigma F}|(u_m)_n) = \text{rk}(R_{\sigma'}|(u_m^s)_n)$, and $R_{\sigma'} = \bigoplus_{\gamma} R_{\gamma}$ with $R_{\gamma} : (u_m^s)_{\gamma} \rightarrow (u_m^s)_{\gamma+\delta}$. Moreover $\text{rk } R_{\gamma} = \text{rk } R_{\rho\gamma}$, and $\text{rk } R_{\gamma} = \text{rk } R_{\theta(\Gamma-\gamma-\delta)}$, where Γ is the weight of the top degree of u_m^s and $\theta(n_0, n_1, n_2) = (n_1, n_0, n_2)$.*

Proof. Ranks are invariant under extension of scalars, ψ carries σF to σ' , and σ' is defined over $\mathbb{F}_2$; σ' has weight δ , and ρ is an automorphism fixing σ' and mapping block γ to $\rho\gamma$. For the duality, u_m^s is a finite-dimensional connected graded Hopf algebra whose top degree is one-dimensional (spanned by the product of all PBW letters, a two-sided integral). Let λ be the projection onto the top degree. The pairing $(a, b) \mapsto \lambda(ab)$ is nondegenerate on $(u_m^s)_{\gamma} \times (u_m^s)_{\Gamma-\gamma}$: u_m^s is Frobenius [7] and, for graded connected algebras, any Frobenius form may be taken to be this projection. Since $\lambda((y\sigma')z) = \lambda(y(\sigma'z))$, the map R_{γ} is adjoint to left multiplication $L_{\sigma'} : (u_m^s)_{\Gamma-\gamma-\delta} \rightarrow (u_m^s)_{\Gamma-\gamma}$. Finally Θ' is an anti-automorphism with $\Theta'(\sigma'z) = \Theta'(z)\sigma'$, mapping weight μ to $\theta(\mu)$, so it conjugates $L_{\sigma'}$ on block μ to $R_{\sigma'}$ on block $\theta(\mu)$. (The duality was also verified on every block for $m \leq 10$.) $\square$

5. COLUMN SUBALGEBRAS AND EXPLICIT KERNELS

For $j \in \mathbb{Z}/3$ put $c_k^{(j)} = e_{k-1+j}\varphi^k$. In A^s one has $c_a c_b = [3 \mid b] c_{a+b}$, hence

$$[c_a, c_b] = ([3 \mid a] + [3 \mid b]) c_{a+b}, \quad c_a^{[2]} = [3 \mid a] c_{2a}.$$

These are the root vectors supported in a fixed column of the loop algebra.

Lemma 5.1. *The restricted subalgebra $H^{(j)}$ generated by c_1, c_2, c_3 is $\text{span}\{c_k : 3 \nmid k\} \oplus \text{span}\{c_{3 \cdot 2^r} : r \geq 0\}$.*

Proof. The right-hand side is closed: a nonzero bracket needs exactly one index divisible by 3, so the sum is not, and a nonzero square needs $3 \mid a = 3 \cdot 2^r$. It is generated: $c_{k+3} = [c_k, c_3]$ for $3 \nmid k$ and $c_{3 \cdot 2^{r+1}} = c_{3 \cdot 2^r}^{[2]}$. $\square$

Thus in $L^s/L_{\geq m}^s$ the image of $H^{(j)}$ has basis $\{c_k : k \in \mathcal{H}_m\}$. The squaring chain $c_3 \rightarrow c_6 \rightarrow c_{12} \rightarrow \dots$ acts as a binary odometer: carries propagate only along the imaginary column vectors of degree $3 \cdot 2^r$, while $c_9, c_{15}, c_{18}, \dots$ are never reached.

Lemma 5.2. *Let $K \subset L^s/L_{\geq m}^s$ be a graded restricted subalgebra with homogeneous basis $z_1, \dots, z_r$, and $\omega_K = z_1 \cdots z_r$. Then $\{y \in u_m^s : yz = 0 \ \forall z \in K\} = u_m^s \omega_K$, of dimension 2^{N_m-r} .*

Proof. By PBW, with a complementary basis ordered before the z_i , $u_m^s \cong V \otimes u(K)$ as right $u(K)$ -modules. So y annihilates K iff each coefficient is a right integral of the finite-dimensional Hopf algebra $u(K)$; the integrals form a one-dimensional space [7], spanned by ω_K (which spans the top degree of $u(K)$). $\square$

Theorem 5.3. *For every $m \geq 2$ and $j \in \mathbb{Z}/3$ let $\omega_j = \prod_{k \in \mathcal{H}_m} c_k^{(j)}$. Then*

$$u_m^s \omega_0 + u_m^s \omega_1 + u_m^s \omega_2 \subset \ker R_{\sigma'}.$$

The ω_j are nonzero of degree $d(m)$ with pairwise distinct Q -weights. Consequently $k_m \leq d(m)$ and the kernel of $R_{\sigma F}$ on $(u_m)_{d(m)}$ has dimension ≥ 3 .

Proof. By Proposition 4.3, $\sigma' \in c_1 u + c_2 u + c_3 u$ (for $j = 0$; apply ρ for $j = 1, 2$). If y annihilates c_1, c_2, c_3 on the right, it annihilates the restricted subalgebra they generate, which is $H^{(j)}$ by Lemma 5.1, and then $y\sigma' = 0$. Lemma 5.2 identifies these y . The weight of $c_k^{(0)}$ has $n_0 - n_2 = 1$ if $3 \nmid k$ and 0 otherwise, so ω_0 is not rotation invariant. $\square$

Remark 5.4. $d(m) = m(m-1)/2$ for $m \leq 9$, but $d(10) = d(9) = 36$, $d(11) = 46$, and $d(m) = m^2/3 + O(m \log m)$; see Figure 1.

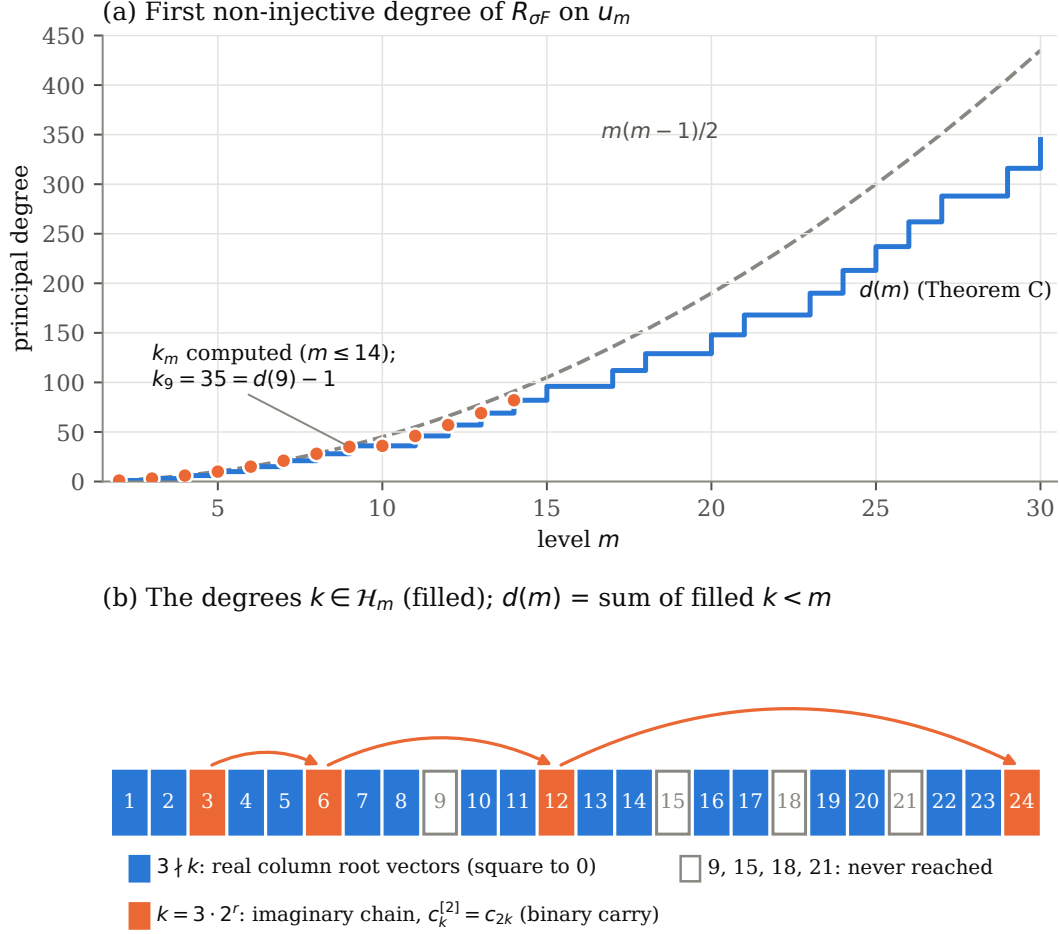


FIGURE 1. (a) The first non-injective degree k_m of $R_{\sigma F}$ on u_m (dots, computed for $m \leq 14$; $k_{15} \leq 95$) against $d(m)$ of Theorem C (steps) and the naive guess $m(m-1)/2$ (dashed). (b) The mechanism: the column subalgebra generated by c_1, c_2, c_3 contains c_k for $3 \nmid k$ and the imaginary chain $c_3 \rightarrow c_6 \rightarrow c_{12} \rightarrow c_{24}$ produced by the 2-map (a binary carry), but never $c_9, c_{15}, c_{18}, c_{21}, \dots$; the staircase $\omega = \prod_{k \in \mathcal{H}_m} c_k$ has degree $d(m)$ and is annihilated by σF (Theorem 5.3).

6. THE CENTRAL FILTRATION

Proposition 6.1. L_m is central in $L/L_{\geq m+1}$ with zero 2-map, u_{m+1} is a free module over $\Lambda(L_m)$ with $u_{m+1}/L_m u_{m+1} = u_m$, and for every degree (and, in the split model, every Q -block γ)

$$\dim \ker R_{\gamma}^{(m+1)} \leq \sum_S \dim \ker R_{\gamma - \text{wt}(S)}^{(m)},$$

the sum over subsets S of a homogeneous basis of L_m^s . Consequently

$$k_{m+1} \geq k_m \quad \text{and} \quad \dim \ker R^{(m+1)} / |G_{m+1}| \leq \dim \ker R^{(m)} / |G_m|.$$

Proof. $[L_i, L_m] \subset L_{\geq m+1}$ and $L_m^{[2]} \subset L_{2m}$. With $J = L_m u_{m+1}$, $J^i / J^{i+1} \cong \Lambda^i(L_m) \otimes u_m$ and $R_{\sigma F}$ preserves the J -adic filtration with $\text{gr } R = 1 \otimes R^{(m)}$; apply $\dim \ker f \leq \dim \ker \text{gr } f$. $\square$

7. σF IS A NON-ZERO-DIVISOR

Although $u(L)$ is not a domain (Theorem 3.1), the leading form of F is regular.

Theorem 7.1. *Left and right multiplication by σ' are injective on $U = u(\widehat{\mathfrak{n}}) = u(L^s)$. Consequently σF is a two-sided non-zero-divisor in $u(L) = \text{gr } \mathbb{F}_2[[P]]$, F is a two-sided non-zero-divisor in $\mathbb{F}_2[[P]]$, and $k_m \geq m - 3$ for all m ; in particular $k_m \geq \max(57, m - 3)$ for $m \geq 12$.*

Proof. (1) With $e_i = y_{1,i}$ the three Chevalley generators one checks $\sigma' = e_0 e_1 e_2 + e_1 e_2 e_0 + e_2 e_0 e_1$. (2) Let $\rho^*(x) = (\pi(x) + \text{tr } \pi(x) I)^T$, where π is the natural representation of Proposition 4.2; this is a graded restricted representation of L^s over $\mathbb{F}_2[t]$, injective on L^s , with $e_0 \mapsto tE_{20}$, $e_1 \mapsto E_{01}$, $e_2 \mapsto E_{12}$, so that $\rho^*(\sigma') = t \cdot I$. (3) Let Ψ_r be the action of U (via its coproduct) on $V^*(t_1) \otimes \cdots \otimes V^*(t_r)$, with independent variables t_i . Specialising $t_r = 0$ makes $\Psi_r(\sigma')$ block upper triangular, and induction gives $\det \Psi_r(\sigma')|_{(t,0,\dots,0)} = t^{3^r}$; hence $\det \Psi_r(\sigma') \neq 0$ and $\Psi_r(\sigma')$ is invertible over $\mathbb{F}_2(t_1, \dots, t_r)$. (4) The matrix coefficients of ρ^* generate the graded dual U^* as an algebra: by graded Nakayama it suffices that they span the indecomposables $Q(U^*) \cong P(U)^* = (L^s)^*$, which holds because ρ^* is injective on L^s . Thus for every $y \neq 0$ there is r with $\Psi_r(y) \neq 0$, and $\Psi_r(y\sigma') = \Psi_r(y)\Psi_r(\sigma') \neq 0$; similarly on the left. The statement for σF follows by Corollary 4.4 (injectivity is invariant under $\otimes \mathbb{F}_8$). Since $\mathbb{F}_2[[P]]$ is complete for the filtration with $\text{gr } \mathbb{F}_2[[P]] = u(L)$, a non-zero-divisor leading form forces F to be a non-zero-divisor. Finally, u_m and U agree in degrees $< m$, so $R_{\sigma'}$ is injective on $(u_m)_n$ for $n + 3 < m$, i.e. $k_m \geq m - 3$. Each step was also checked by machine; a more detailed write-up of (3) and (4) is in `results/THEORY.md`, §7.1, of the repository. $\square$

Remark 7.2. Theorem 7.1 is the “Kaplansky part” of the expected statement $\text{rk}_P(F) = 1$; it does not by itself give $c_e = 0$, because $\mathbb{F}_2[[P]]$ is not known to embed in a division ring. The method gives only a linear bound for k_m : on u_m one would need $\Psi_r(\sigma')^{-1}$ to have monomial denominators, but already $\det \Psi_3(\sigma') = s_1^{21}(s_1^3 + t_1 t_2 t_3)^2$ with $s_1 = t_1 + t_2 + t_3$. A natural route to a quadratic bound is to show that every kernel element at level m dies at level $m - a$; by Frobenius duality this is equivalent to $\omega_{K_a} \in u_m \sigma'$, where ω_{K_a} is the product of all letters of degree $m - a, \dots, m - 1$, and it holds for $a = 3, 6 \leq m \leq 10$. However, this route cannot succeed: if $T_a(m) = \sum_{k=m-a}^{m-1} k d_k \leq k_{m-a} + 2$, then $R_{\sigma'}$ is onto near the top of the ideal generated by $L_{\geq m-a}$ and the integral of u_{m-a} lifts to a kernel element at level m , so $\omega_{K_a} \notin u_m \sigma'$. Since $T_a(m) < 3am$ while $k_{m-a} \geq m - a - 3$ grows, for every fixed a the criterion fails for infinitely many m , and any chain of such steps yields at most $k_m \leq b + 4m^{3/2}$, which is just short of what Theorem 9.4 needs. A quadratic bound is equivalent to the vanishing of $H_1(L_{\geq m}; U/U\sigma')$ below degree cm^2 , since $\ker(R^{(m)})_n \cong \text{Tor}_1^{u(L_{\geq m})}(\mathbb{F}_2, U/U\sigma')_{n+3}$.

8. COMPUTATIONS

The bounds of the table are plotted in Figure 2.

Remark 8.1. The kernel at degree k_m is 3-dimensional and spanned by $\omega_0, \omega_1, \omega_2$ for $m \in \{2, \dots, 8\} \cup \{10, \dots, 14\}$. The case $m = 9$ is exceptional: $k_9 = 35 = d(9) - 1$, and the kernel there is spanned by the rotations of a “lowered staircase” η , which is not annihilated by any single letter and arises from a cancellation inside $\eta\sigma'$; it leaves the kernel at $m = 10$. In particular the empirical rule $k_m = m(m - 1)/2$, valid for $m \leq 8$, fails for $m \geq 9$.

9. MAXIMAL-RANK BASELINE AND EXCESS

Throughout, $h_n = \dim(u_m)_n$, top is the top degree of u_m , $\text{mid} = \text{top}/2$, and $\ker_n = \dim \ker(R_{\sigma'}|_{(u_m)_n})$. Let X_m be the degree of a uniformly random PBW monomial of u_m ,

m	$ G_m $	$\dim C_m$ (exact)	graded cokernel	bound on c_e	k_m
2	2^3	7	7	0.29167	1
3	2^6	37	37	0.19271	3
4	2^8	100	100	0.13021	6
5	2^{11}	547	547	0.08903	10
6	2^{14}	3170	3182	0.06449	15
7	2^{16}	10226	10226	0.05201	21
8	2^{19}	—	64216	0.04083	28
9	2^{22}	—	423412	0.03365	35
10	2^{24}	—	1526176	0.03032	36
11	2^{27}	—	—	—	46
12	2^{30}	—	—	—	57
13	2^{32}	—	—	—	69
14	2^{35}	—	—	—	82
15	2^{38}	—	—	—	$\leq$ 95

TABLE 1. Exact data. Values for $m \leq 6$ reproduce [1, Table 6.1]; bold entries are new. The bound column is $\dim C_m/n_m$ for $m \leq 7$ and (graded cokernel)/ n_m for $m \geq 8$ (Lemma 2.4). k_m is the first degree where $R_{\sigma F}$ fails to be injective on u_m .

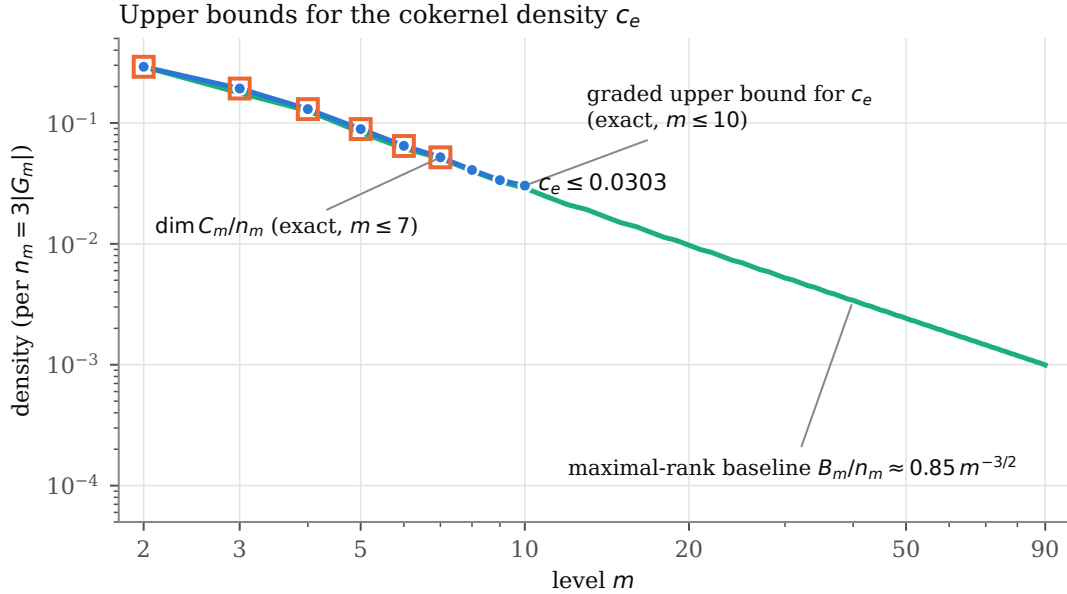


FIGURE 2. Upper bounds for c_e (log-log). Squares: exact $\dim C_m/n_m$ ($m \leq 7$). Dots: the graded upper bound $\dim \text{coker}(R_{\sigma F}|u_m)/n_m$ (Lemma 2.4), exact for $m \leq 10$, giving $c_e \leq 0.0303$. Line: the maximal-rank baseline B_m/n_m (Section 9), computed exactly for $m \leq 90$; the graded bound equals the baseline plus the excess $e_m/3$, and $c_e = 0$ is equivalent (at the graded level) to $e_m \rightarrow 0$.

i.e. $X_m = \sum_{k < m} \sum_{i=1}^{d_k} k \varepsilon_{k,i}$ with independent fair bits $\varepsilon_{k,i}$, so that $h_n = |G_m| \Pr[X_m = n]$ and $\mathbb{E}X_m = \text{mid}$.

Trivially $\dim \ker R_\gamma \geq \max(0, \dim u_\gamma - \dim u_{\gamma+\delta})$. Put $B_m^Q = \sum_\gamma \max(0, \dim u_\gamma - \dim u_{\gamma+\delta})$, and define the excess e_m by

$$x_m := \frac{\dim \ker R_{\sigma F}|u_m}{|G_m|} = \frac{B_m^Q}{|G_m|} + e_m, \quad c_e \leq x_m/3.$$

Lemma 9.1. *There is an absolute constant A with $\max_n \Pr[X_m = n] \leq A m^{-1/2}$ and $B_m := \sum_n \max(0, h_n - h_{n+3}) \leq A m^{-1/2} |G_m|$.*

Proof. For each $k < m - 3$ with $k \equiv 1, 2 \pmod{6}$ pick one letter of degree k and one of degree $k + 3$; these $M \geq m/3 - 2$ pairs are disjoint. Let T be the number of pairs in which exactly one of the two letters occurs. Conditionally on T and on all other bits, $X_m = Y + 3 \text{Bin}(T, \frac{1}{2})$ with Y fixed. Hence $\max_n \Pr[X_m = n]$ and the total variation distance $d_{TV}(X_m, X_m + 3)$ are bounded by $\mathbb{E}[\min(1, A_0 T^{-1/2})]$, and $T \sim \text{Bin}(M, \frac{1}{2})$ gives $O(M^{-1/2})$. Finally $\sum_n (h_n - h_{n+3}) = 0$, so $B_m = \frac{1}{2} \sum_n |h_n - h_{n+3}| = |G_m| d_{TV}(X_m, X_m + 3)$. $\square$

Since B_m^Q and B_m agree numerically to four digits, Lemma 9.1 shows rigorously that the degree-wise baseline is $O(m^{-1/2})$; the exact values for $m \leq 90$ follow $B_m/|G_m| \approx 2.55 m^{-3/2}$, as a local central limit theorem predicts ($\text{Var } X_m = \frac{1}{4} \sum_{k < m} k^2 d_k \sim 2m^3/9$). The excess is small: $e_m = 0.0025, 0.0022, 0.0027, 0.0041$ for $m = 7, \dots, 10$, symmetric about the middle degree.

Conjecture 9.2. $e_m \rightarrow 0$. *Equivalently (given the baseline asymptotics) the graded cokernel density tends to 0, and hence $c_e = 0$.*

Conjecture 9.2 is a Lefschetz-type maximal-rank statement for multiplication by the weight- δ element σ' on the truncated restricted enveloping algebras of $\hat{\mathfrak{n}}$; compare [9] for the commutative theory. The exceptional element η at $m = 9$ shows that a proof cannot rest on annihilators alone.

9.1. The excess is a special-position effect. The weight- δ subspace of u_m^s is 6-dimensional (spanned by $y_{3,0}, y_{3,1}, y_{2,q}y_{1,q+1}$ and $y_{1,0}y_{1,1}y_{1,2}$), so all 63 nonzero $\mathbb{F}_2$ -elements can be tested.

Proposition 9.3 (proof by computation). *For $4 \leq m \leq 8$, a generic element of $(u_m^s)_\delta$ (over $\overline{\mathbb{F}_2}$) has maximal rank on every Q -block, i.e. excess 0. For $m = 9$ its excess is at most 18, concentrated in 18 small square blocks near the middle degree. By contrast σ' has excess 3, 30, 156, 162, 1136, 11200 for $m = 4, \dots, 9$.*

(Maximal rank is certified blockwise by exhibiting one element, over $\mathbb{F}_2, \mathbb{F}_{16}$ or $\mathbb{F}_{256}$, of maximal rank; rank is lower semicontinuous.) Thus the truncated algebras u_m^s have the maximal-rank property for weight δ almost perfectly, and the excess of σ' comes from its special position. Semicontinuity bounds $\text{rk } R_{\sigma'}$ in the wrong direction, so this does not by itself prove Conjecture 9.2.

9.2. A conditional proof.

Theorem 9.4. *Suppose that for some constant C and infinitely many m :*

- (H1) $\ker_n \leq C h_{n-k_m}$ for $k_m \leq n \leq \text{mid}$, and
- (H2) $k_m \geq d(m) - 1$.

Then along these m

$$x_m \leq 2C \Pr[X_m \leq \text{mid} - k_m] + 6 \max_n \Pr[X_m = n] \leq 2C e^{-m/4+O(1)} + 6A m^{-1/2},$$

and consequently $c_e = 0$. In (H2) it suffices that $k_m/m^{3/2} \rightarrow \infty$.

Proof. Let r_n be the rank of $R_{\sigma'} : (u_m)_n \rightarrow (u_m)_{n+3}$, so $\ker_n = h_n - r_n$. By the duality of Corollary 4.4 (summed over blocks of a fixed principal degree), $\ker_n = h_{\text{top}-n} - r_{\text{top}-n-3}$. Writing $j = \text{top} - n - 3$ this reads $\ker_n = (h_{j+3} - h_j) + \ker_j$. Summing over $n > \text{mid}$, the terms $h_{j+3} - h_j$ telescope to at most $3 \max_n h_n$, and the degrees n within 3 of mid contribute at most $3 \max_n h_n$ more; hence

$$\dim \ker R_{\sigma'} \leq 2 \sum_{n \leq \text{mid}} \ker_n + 6 \max_n h_n.$$

By (H1), $\sum_{n \leq \text{mid}} \ker_n \leq C \sum_{j \leq \text{mid} - k_m} h_j = C |G_m| \Pr[X_m \leq \text{mid} - k_m]$. Hoeffding's inequality for X_m (a sum of independent variables with ranges k) gives $\Pr[X_m \leq \text{mid} - k_m] \leq \exp(-2k_m^2 / \sum_{k < m} k^2 d_k)$; with $k_m \geq d(m) - 1 \sim m^2/3$ and $\sum_{k < m} k^2 d_k \sim 8m^3/9$ the exponent is $-m/4 + O(1)$, and it tends to $-\infty$ as soon as $k_m/m^{3/2} \rightarrow \infty$. The last term is bounded by Lemma 9.1. Since x_m is non-increasing (Proposition 6.1), a subsequence suffices, and $c_e \leq x_m/3$. $\square$

Hypothesis (H1) holds with $C = 3$ for $5 \leq m \leq 10$ (the ratio $\ker_n / h_{n-k_m}$ equals 3 at $n = k_m$ and decreases afterwards), and (H2) holds for $m \leq 14$: an exact layer-wise lifting from level 10 gives $k_{12} = 57 = d(12)$, $k_{13} = 69 = d(13)$ and $k_{14} = 82 = d(14)$ (all ρ -orbits of blocks were computed; for $m = 14$ the largest block has dimension 16 769 439), and in each case the kernel at $d(m)$ is exactly the span of $\omega_0, \omega_1, \omega_2$. Hence $R_{\sigma F}$ is injective in all degrees ≤ 81 for every $m \geq 14$ (Proposition 6.1). For $m = 15$ we found the predicted exceptional element: an element $\eta_{15} \equiv y_{13,1} \omega_0^{(14)} \pmod{L_{14}}$ of degree $95 = d(15) - 1$ with $\eta_{15} \sigma' = 0$, so $k_{15} \leq 95$; like η at $m = 9$, it is not annihilated by any single letter and it leaves the kernel at $m = 16$. Injectivity below 95 was checked only on the smaller blocks of degrees 94 and 95 (the only lower bound proved is $k_{15} \geq k_{14} = 82$). For $m = 11$ a partial computation (all blocks of dimension $\leq 2 \cdot 10^4$, remaining blocks bounded via Proposition 6.1) gives $0.00051 \leq e_{11} \leq 0.01693$.

Remark 9.5. The condition $k_m/m^{3/2} \rightarrow \infty$ is sharp for this method: if $k_m = O(m^{3/2})$, the central limit theorem keeps $\Pr[X_m \leq \text{mid} - k_m]$ bounded below. One can also show that $k_m \rightarrow \infty$ if and only if $R_{\sigma'}$ is injective on the untruncated algebra $u(\hat{\mathfrak{n}})$, in which case $k_m \geq m - 3$; otherwise k_m is eventually constant (≥ 57). Moreover, the column annihilators of Section 5 cannot by themselves prove (H2): from $m = 7$ on, the dual witnesses for the disappearance of ω_j at a jump step are of cancellation type.

10. GRADED ZERO DIVISORS DO NOT LIFT

Proposition 10.1 (proof by computation). (1) *For every $0 \neq v \in V$ and every $x \in \mathbb{F}_2[[P]]$ with leading form v , $x^2 \in I^9 \setminus I^{10}$. The obstruction is a nonzero class in $\ker \text{ad}_v / \text{im ad}_v$ in degree 9.*

(2) *For every pair $(v, w) \in (V \setminus 0)^2$ and all lifts $x, y, xy \notin I^{11}$.*

(3) *For the (3, 14) family of the Remark in Section 3, $xy \in I^{18} \setminus I^{19}$ for all lifts.*

These statements are exact: they solve the lifting equations as linear systems in $\text{gr } \mathbb{F}_2[[P]]$, enumerating the full affine solution space where the equations become nonlinear (for instance all 2^9 choices in (2)). Moreover, for the lifts $\tilde{v}$ the kernel density of right multiplication on $\mathbb{F}_2[G_m]$ is 0.375, 0.279, 0.234 for $m = 5, 6, 7$, decreasing like that of a generic element of I^4 (0.332, 0.244, 0.203), whereas in the graded algebra it stays at 0.625. This is consistent with the pro-2 Kaplansky conjecture for P , but of course does not prove it.

11. OTHER CYCLIC ALGEBRAS

The construction of Section 4 makes sense for every prime p and every degree $\ell \geq 2$ (we write ℓ for the degree, to avoid a clash with $d(m)$ and d_k). Let $A_{p,\ell}^s = \mathbb{F}_p^\ell[\varphi; s]$ with s the cyclic shift of the idempotents $e_0, \dots, e_{\ell-1}$, let $t = (1, \dots, 1)\varphi^\ell$, and put $L_{p,\ell} = (A_{p,\ell}^s)_{>0}/t\mathbb{F}_p[t]$ with the commutator bracket and the p -th power map. Exactly as in Proposition 4.1, $L_{p,\ell} \otimes \mathbb{F}_{p^\ell} \cong \text{gr } P_{p,\ell} \otimes \mathbb{F}_{p^\ell}$ for the pro- p group $P_{p,\ell} = (1 + \varphi\mathcal{O})/(1 + t\mathbb{F}_p[[t]])$, $\mathcal{O} = \mathbb{F}_{p^\ell}[[\varphi; \sigma]]$, $\varphi^\ell = t$, and for $p \nmid \ell$ the algebra $L_{p,\ell}$ is the positive part of $A_{\ell-1}^{(1)}$ in its principal grading. Write u_m, k_m as before, with $y_{k,i} = e_i\varphi^k$, and let

$$\sigma'_{p,\ell} = \sum_{r \in \mathbb{Z}/\ell} e_r e_{r+1} \cdots e_{r+\ell-1}, \quad e_i = y_{1,i},$$

the cyclic sum of the Coxeter word. For $(p, \ell) = (2, 3)$ this is the leading form σ' of F (step (1) of the proof of Theorem 7.1). We have not constructed lattices and Hecke elements for other (p, ℓ) ; the results of this section concern the graded model with $\sigma'_{p,\ell}$ as the natural candidate for the leading form. In this generality the analogue of Theorem B holds verbatim, the non-zero-divisor theorem extends to all $p \nmid \ell$, and the explicit kernels of Theorem C do *not* extend, except for $\ell = 2$. All computations of this section use an independent implementation for general p (Appendix A(f)).

Theorem 11.1. *If $p \nmid \ell$, then left and right multiplication by $\sigma'_{p,\ell}$ are injective on $u(L_{p,\ell})$; hence $k_m \geq m - \ell$ for all m .*

Proof. The proof of Theorem 7.1 goes through with two changes. First, with $\pi(e_i) = E_{ii}$, $\pi(\varphi) = \sum_i E_{i+1,i} + tE_{0,\ell-1}$, the map $\rho(x) = \pi(x) - \ell^{-1} \text{tr } \pi(x) I$ kills $t\mathbb{F}_p[t]$ and is a restricted representation, because $(X - cI)^p = X^p - c^p I$, $\text{tr}(X^p) = (\text{tr } X)^p$ in characteristic p , and $\ell^p \equiv \ell$; this is where $p \nmid \ell$ is used. Second, in $\rho^*(x) = -\rho(x)^T$ one has $\rho^*(e_i) = -E_{i-1,i}$ for $i \neq 0$ and $\rho^*(e_0) = -tE_{\ell-1,0}$, so each cyclic word maps to $(-1)^\ell t E_{r-1,r-1}$ and $\rho^*(\sigma'_{p,\ell}) = (-1)^\ell t I$; the triangularity argument then gives $\det \Psi_r(\sigma'_{p,\ell})|_{(t,0,\dots,0)} = \pm t^{\ell^r}$. Faithfulness is unchanged. $\square$

Remark 11.2. For $p \mid \ell$ the scalar loops are not split off and the argument fails. For $(p, \ell) = (2, 2)$ one has $\sigma' = e_0 e_1 + e_1 e_0 = [e_0, e_1] = t = 0$, and for $(3, 3)$ the Coxeter term $y_{1,0} y_{1,1} y_{1,2}$ of σ' has coefficient $3 = 0$. Nevertheless σ' is injective on $u(L_{p,\ell})$ in all degrees ≤ 9 for $(3, 3)$ and ≤ 8 for $(2, 4)$ (exact computation).

11.1. The column mechanism is a characteristic-2 phenomenon. Theorem 5.3 rests on $\sigma' \in c_1 u + c_2 u + c_3 u$ for the column letters $c_k = e_{k-1} \varphi^k$. The same argument works for any (p, ℓ) , with $c_1, \dots, c_\ell$, as soon as $\sigma'_{p,\ell}$ lies in the right ideal $c_1 u + \dots + c_\ell u$; the column integrals are then kernel elements in degree $(p-1)(\sum_{k < m, \ell \nmid k} k + \sum_{\ell p^r < m} \ell p^r)$.

Proposition 11.3 (proof by computation). *$\sigma'_{p,\ell}$ lies in a column ideal (equivalently, by the rotation symmetry, in all of them) for $(p, \ell) = (2, 3)$ and for $\ell = 2, p = 3, 5, 7$. It lies in no column ideal and in no row ideal (letters $e_j \varphi^k$, j fixed), for either orientation of the Coxeter word, for*

$$(p, \ell) \in \{(5, 3), (7, 3), (3, 3), (2, 4), (3, 4), (2, 5), (3, 5), (2, 7)\}.$$

For $\ell = 2$ the column ideal contains the whole weight space u_δ . Over $\mathbb{Z}$ (computed modulo a large prime and lifted), the class of σ'_ℓ modulo the column ideal in weight δ is 0 for $\ell = 2$, is $2(y_{3,0} - y_{2,0} y_{1,1})$ for $\ell = 3$, and has coprime integer coordinates for $\ell = 4, 5, 6$.

Thus for $\ell = 3$ the explicit kernels of Theorem 5.3 exist only because of the factor 2 in this class, and for $\ell \geq 4$ they do not exist for any $p > \ell$ nor for the small p tested. In the cases without the column mechanism the first kernel lies elsewhere: for $(5, 3)$ one finds $k_3 = 14 > 12$.

11.2. **The case $\ell = 2$, p odd.** Put $x_k = y_{k,0}$, $f_k = y_{k,1}$ (k odd) and $h_k = y_{k,0}$ (k even). Then $[x_i, f_j] = 2h_{i+j}$, $[h_j, x_i] = x_{i+j}$, $[h_j, f_i] = -f_{i+j}$, all other brackets of letters vanish, $x_k^{[p]} = f_k^{[p]} = 0$, $h_j^{[p]} = h_{pj}$, and $\sigma' = x_1 f_1 + f_1 x_1 = 2x_1 f_1 - 2h_2$: the loop algebra of $\mathfrak{sl}_2$ in its principal grading, and σ' is the anticommutator of the two Chevalley generators.

Theorem 11.4. *Let p be odd, $\ell = 2$, $m \geq 3$.*

- (1) $H = \text{span}\{x_k : k \text{ odd}\} \oplus \text{span}\{h_{2p^r}\}$ is a restricted subalgebra with $\sigma' \in Hu$, so its integral ω_H lies in $\ker R_{\sigma'}$, in degree $\text{od}_p(m) = (p-1)(\sum_{k < m \text{ odd}} k + \sum_{2p^r < m} 2p^r)$.
- (2) $K = \text{span}\{x_k \ (k \equiv 1), f_k \ (k \equiv 3), h_k \ (k \equiv 0 \pmod{4})\}$, the image of the loop algebra under $t \mapsto t^2$, is a restricted subalgebra, and for $0 \leq j < p$

$$f_1^j \sigma' \equiv -2(2j+1) h_2 f_1^j \pmod{Ku_m}.$$

Hence $\omega_K f_1^j \in \ker R_{\sigma'}$ if and only if $j = (p-1)/2$, in degree

$$\kappa_p(m) = (p-1) \sum_{k < m, k \not\equiv 2 \pmod{4}} k + \frac{p-1}{2}.$$

Consequently $k_m \leq \min(\text{od}_p(m), \kappa_p(m))$.

Proof. For a restricted subalgebra K let ω_K be the product of the $(p-1)$ -st powers of a homogeneous basis; as in Lemma 5.2 it spans the right integrals of $u(K)$, so $\omega_K k = 0$ for $k \in K$. Hence $zv \in Ku_m$ implies $\omega_K zv = 0$. By PBW $u_m = u(K) \otimes V$ and $u_m/Ku_m \cong V$ has the PBW monomials in the letters outside K as a basis. (1) Both terms of $2x_1 f_1 - 2h_2$ begin with a letter of H . (2) Closure of K : $[x_{4a+1}, f_{4b+3}] = 2h_{4(a+b+1)}$, $[h_{4a}, x_{4b+1}] = x_{4(a+b)+1}$, $[h_{4a}, f_{4b+3}] = -f_{4(a+b)+3}$, $h_{4a}^{[p]} = h_{4ap}$. Since $f_3 \in K$ commutes with f_1 and $[h_2, f_1] = -f_3$, one has $f_1^a h_2 f_1^b \equiv f_1^{a+b} h_2 \equiv h_2 f_1^{a+b}$ modulo Ku_m . Expanding $[x_1, f_1^j] = 2 \sum_{a+b=j-1} f_1^a h_2 f_1^b$ gives $f_1^j x_1 f_1 \equiv -2j h_2 f_1^j$ and $f_1^{j+1} x_1 \equiv -2(j+1) h_2 f_1^j$, because $x_1(\dots) \in Ku_m$. The sum is $-2(2j+1) h_2 f_1^j$, and $h_2 f_1^j$ is a basis vector of V . $\square$

Both families were also checked by direct multiplication in u_m ($p = 3$: $m \leq 16$; $p = 5$: $m \leq 12$; $p = 7$: $m \leq 10$), including the uniqueness of the exponent $(p-1)/2$. At $p = 2$ the coefficient $-2(2j+1)$ vanishes identically, in accordance with $\sigma'_{2,2} = 0$. Asymptotically $\text{od}_p(m) \sim (p-1)m^2/4 < \kappa_p(m) \sim 3(p-1)m^2/8$, so the second family matters only for small m .

p	m	2	3	4	5	6	7	8	9	10	11
3	$\min(\text{od}, \kappa)$	2	3	9	12	22	27	41	48	66	66
3	k_m	2	3	9	12	22	27	41	48	64	66
5	$\min(\text{od}, \kappa)$	4	6	18	24	44	44				
5	k_m	4	6	18	24	42	44				
7	$\min(\text{od}, \kappa)$	6	9	27	36	66					
7	k_m	6	9	26	34	≥ 63					

TABLE 2. Thresholds for $\ell = 2$ (exact). Bold: levels where k_m is below the bound of Theorem 11.4.

At every level of Table 2 without a bold entry, the kernel in degree k_m is spanned by the elements of Theorem 11.4 and their images under the symmetry $x \leftrightarrow f$. The bold entries are explained by cancellation-type elements of the kind met at $m = 9, 15$ in Section 9: at $p = 3$, $m = 10$ the kernel in degree 64 is spanned by a 24-term lowering of ω_H and its mirror image.

In every case the defect is at most 2 and the element dies at the next level. For $p = 3$, $m = 11$ this gives $k_{11} = 66$ by the following window argument. If $n < k_a + a$, the reduction $u_m \rightarrow u_a$ is injective on $\ker(R_{\sigma'}|(u_m)_n)$, block by block: filter u_m by the powers of the right ideal generated by $L_{\geq a}/L_{\geq m}$ as in the proof of Proposition 6.1; a kernel element in the i -th step, $i \geq 1$, would have a leading part in $\text{gr}^i u(L_{\geq a}) \otimes \ker R^{(a)}$ of u_a -degree $\leq n - a < k_a$. With $a = 10$, $k_{10} = 64$ it suffices to check the six blocks that carry the level-10 kernel in degrees 64 and 65; all six have zero kernel at level 11, and ω_H gives $k_{11} \leq 66$.

The density behaves differently from the case (2, 3). Splitting $x_m = \dim \ker R_{\sigma'}/\dim u_m$ into the degree baseline $\sum_n \max(0, h_n - h_{n+2})/\dim u_m$ and an excess (this only overestimates the excess e_m of Section 9, which uses the block baseline B_m^Q), one finds:

case	levels	excess
$\ell = 2, p = 3$	$m = 4, \dots, 9$	0.0082, 0.0082, 0.0021, 0.00081, 0.00012, 0.00006
$\ell = 2, p = 5$	$m = 4, 5, 6$	0.0026, 0.0015, 0.00033
$(2, 3), e_m$	$m = 7, \dots, 10$	0.0025, 0.0022, 0.0027, 0.0041

The baseline is $O(m^{-1/2})$ by Lemma 9.1, whose proof works with shift 2. So the graded analogue of c_e vanishes for $\ell = 2$ if and only if the excess tends to 0, and the data support this much more strongly than for (2, 3). Theorem 9.4 transfers verbatim with $\deg \sigma' = 2$: (H1) together with $k_m \geq \min(\text{od}_p(m), \kappa_p(m)) - O(1)$, or merely $k_m/m^{3/2} \rightarrow \infty$, implies that the graded cokernel density tends to 0.

12. QUESTIONS

Question 12.1. Prove Conjecture 9.2, or directly that $\text{rk}_P(F) = 1$. By Theorem 9.4 it suffices to prove (H1) and (H2): the kernel of $R_{\sigma'}$ below the middle degree grows no faster than a fixed multiple of the free module generated in degree k_m , and the threshold stays within one of $d(m)$.

Question 12.2. Is $k_m \in \{d(m)-1, d(m)\}$ for all m ? More precisely, is it true that $k_m = d(m)-1$ if $3 \mid m$, $m \geq 9$ and $m \neq 3 \cdot 2^r$, and otherwise $k_m = d(m)$ with kernel spanned by $\omega_0, \omega_1, \omega_2$? This was formulated from the data for $m \leq 12$ and then confirmed for $m = 13, 14$ ($k_{13} = 69$, $k_{14} = 82$); for $m = 15$ the predicted exceptional kernel element in degree 95 exists.

Question 12.3. Show that $\text{Tor}_1^{u(L_{\geq m})}(\mathbb{F}_2, u(\widehat{\mathfrak{n}})/u(\widehat{\mathfrak{n}})\sigma')$ vanishes in degrees below cm^2 for some $c > 0$ (or below $m^{3/2}\psi(m)$ with $\psi \rightarrow \infty$). This is equivalent to the threshold bound needed in (H2), and by Remark 7.2 it cannot be obtained by iterating level-by-level comparisons.

Question 12.4. Is there a filtration of $\mathbb{F}_2[[P]]$ (for instance a weight filtration adapted to the twisted form of $\widehat{\mathfrak{n}}$) whose associated graded ring is a domain? By Proposition 2.2 such a ring would have subexponential growth, hence be Ore; if moreover $\mathbb{F}_2[[P]]$ then embeds continuously into a division ring Q with a discrete valuation, [4, Thm. 7.1] ($\varphi^\# \text{rk}_Q \leq \text{rk}_P$) would give $\text{rk}_P(x) = 1$ for all $x \neq 0$, in particular $c_e = 0$.

Question 12.5. (a) Is $\sigma'_{p,\ell}$ a non-zero-divisor when $p \mid \ell$ (Remark after Theorem 11.1)? (b) For $\ell = 2$, p odd: is $k_m \geq \min(\text{od}_p(m), \kappa_p(m)) - 2$ for all m , and does the excess tend to 0? Here $\mathfrak{sl}_2$ -representation theory might replace the evaluation-module arguments of Section 7. (c) For $\ell \geq 3$, $(p, \ell) \neq (2, 3)$: which mechanism produces the first kernel of $R_{\sigma'_{p,\ell}}$? (d) Lattices acting simply transitively on the vertices of $\tilde{A}_{\ell-1}$ -buildings are obtained from cyclic division algebras over $\mathbb{F}_q(t)$ [10]. Is the leading form of the corresponding Hecke element (the analogue of F , now $1 - \bar{f} \cdots \bar{f}$ since the Laplacian is $-I + A$ modulo p) the cyclic Coxeter sum $\sigma'_{p,\ell}$?

APPENDIX A. COMPUTATIONAL METHODS

All code, data and logs are in the repository cited on the title page (directories `code/`, `data/` and `results/`); the figures are produced by the script `make_data_and_figures.py`. The group G_m is realised as normalised truncated skew power series in $\mathbb{F}_8[[\varphi; \sigma]]/(\varphi^m)$, with conventions ported line by line from Chen's published scripts [1]. Ranks over $\mathbb{F}_2$ use a bit-packed elimination with 64-column panels and Four-Russians tables (numba). Independent checks: (a) $\dim C_m$ for $m \leq 6$ reproduces [1]; (b) $\dim C_7$ was computed from three matrices of provably equal rank (transpose, left multiplication, random row and column permutations); (c) $u(L)$ was implemented twice (the $\mathbb{F}_8$ -model and the split model, with independent straightening codes) and checked against Jennings products in $\mathbb{F}_2[G_N]$ for $N \leq 12$; (d) the zero divisor of Theorem 3.1 was confirmed by plain group multiplication in $\mathbb{F}_2[G_9]$; (e) the graded cokernels for $m = 9, 10$ use the block decomposition of Corollary 4.4; for $m = 10$ one block per rotation orbit was computed and the remaining blocks were spot-checked; (f) Section 11 uses a second, independent engine for general p and ℓ (directory `code/family/`): monomials with exponents $< p$ encoded in base p , a compiled rewriting system with coefficients, and rank computations modulo p . A pure-Python version and a compiled version agree with each other on 21 test cases and reproduce the (2, 3) values of Table 1 for $m \leq 7$ (kernel dimensions and thresholds).

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